MARK SCHEME · TEACHER COPY

11BR · Topic 7: Ethics and Ownership

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Total marks: 59

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · May/June 2023 Paper 12, original Q7 (d) [3 marks]

Question	Answer	Marks
1(a)	**1 mark** each to **max 3** • Uses speech recognition • ... which identifies key phrases / words spoken • … and matches these to a database • ... and generates the most likely sentence / command / word	3

Question 2 · May/June 2023 Paper 13, original Q5 (b) [4 marks]

Question	Answer	Marks
2(a)	**1 mark** for appropriate licence; **1 mark** for each point to **max 3** • Commercial software licence • User has to pay for the product so the programmer can gain an income • Enables the program to be **copyrighted** • ... so the user cannot **legally** edit the program // the programmer retains control over product • … and can take legal action against people who attempt to **illegally** copy it /sell it on • Shareware licence • Enables the program to be **copyrighted** • The user cannot **legally** edit the program so the developer retains control over product • User can try the program for free and then pay for the full game which allows the programmer to gain an income • so more people can experience it and therefore be more likely to buy it	4

Question 3 · May/June 2024 Paper 13, original Q7 (b) [3 marks]

Question	Answer	Marks
3(a)	**1 mark** each to **max 3**: e.g. • Voice/speech recognition is used • ... to identify if someone speaking • The sound is recorded and analysed • The audio recordings are compared to a database of words/sound waves • ... to identify the word that has the highest probability of being said • Natural language recognition is used • Words are combined and compared to known sentences • … programmed action(s) for matching sentence(s) are performed	**3**

Question 4 · May/June 2025 Paper 12, original Q3 (a),(b),(c) [12 marks]

Question	Answer	Marks

4(a)	**1 mark** each to **max 4** e.g. • Uses image recognition // optical character recognition • to analyse a photograph/pixels to identify the location of characters within an image • in order to convert the pattern of pixels into individual characters • Uses natural language • to combine the characters into words • and compare the words to database to identify a match in the language of the writing • it may also predict what is missing / incomplete / from what has been read • then compare the words to a database of the user's chosen language to find a match and translate • Uses text-to-speech • to generate the audio waveforms corresponding to the words	**4**
4(b)	**1 mark** each to **max 2** e.g. • It can help people with visual impairment to identify which products to purchase • It can help people who cannot understand the language • It can help to overcome learning / reading difficulties	**2**

4(c)(i)	**1 mark** each to **max 3** • The program is usually sold for a fee • Users have no access to source code • It protects the intellectual rights of the programmer / developer • Users cannot legally alter or share the program • There are limitations on the use of the software e.g. number of times it can be installed • Support and maintenance is usually provided e.g. updates, bug fixes etc.	**3**
4(c)(ii)	**1 mark** each to **max 3** e.g. • The source code would be accessible therefore it could be changed • which might lead to the program outputting an incorrect response • so people could purchase the wrong item • which might have serious consequences // by example e.g. allergies	**3**

Question 5 · May/June 2025 Paper 13, original Q3 (d) [3 marks]

Question	Answer	Marks

5(a)	**1 mark** each to **max 3** e.g. • There is no / limited access to legal advice • in case action is taken against them • There are fewer networking opportunities • so they could miss out on contacts / jobs • There would be less access to training • There would be no clear laid out ethical guidelines • and / or people to discuss potential ethical problems with • possibly leading to inappropriate / unethical actions • which might lead to legal proceedings / recourse	**3**

Question 6 · Oct/Nov 2023 Paper 13, original Q6 (a),(b) [6 marks]

Question	Answer	Marks
6(a)	**1 mark** for each bullet point (**max 2**). • Users are able to trial the program and may offer the programmer constructive **feedback** • **More people** might try the program because the trial is free of charge • Allows bugs to be found and corrected on a wide range of computer system configurations improving the programmer's original program • Users that find the trial useful will buy the program so programmer gets **income** • Allows the program to be **copyrighted** and so protects the programmer's intellectual property rights // no **illegal** modification allowed	**2**

6(b)	**1 mark** for each bullet point (**max 4**). • The programmer has ethical guidelines to follow • …so purchasers of programs know the standards being applied • …so programmer does not have to decide what is ethical or not • Clients know programmer is reputable • …recognition of programmer's skills/knowledge • There may be an entry requirement / exam • … so clients know that the programmer is competent • Professional ethical body provides help and support • Such as legal advice // by appropriate example • Enables programmer to attend the ethical body's training courses • …to keep skills up to date // to develop skills	**4**

Question 7 · Oct/Nov 2024 Paper 12, original Q5 (a),(b) [9 marks]

Question	Answer	Marks

7(a)	**1 mark** for each bullet point **max 2** for public **max 2** for colleagues e.g. Colleagues: The programmer should: • Treat colleagues fairly and avoid acts of discrimination • Be prepared to accept / offer critique • … so that technical work can be improved • Credit the contributions of colleagues • … so that they feel valued / respected • Help / train colleagues etc. e.g. The public: The programmer should: • Maintain the health, safety and welfare of the public • … so as not to endanger anyone • Be honest / realistic in making claims about the software • … so that public trust is maintained • Maintain the security of the public's data etc.	**4**

7(b)(i)	Type of software licence **1 mark** for • Open Source Initiative // Free Software Foundation Reason for choice: **1 mark** for each bullet point (**max 2**) • Allows program source code to be examined • … so that the business can adapt/modify the program to meet their needs • Allows the programmer to earn money for the software and updates	**3**
7(b)(ii)	**1 mark** for each bullet point (**max 2**) • To identify themselves as the owner/author // To gain formal recognition of ownership • To allow for legal consequences if anyone steals/copies it • To restrict competition	**2**

Question 8 · Oct/Nov 2024 Paper 13, original Q5 (a),(b) [6 marks]

Question	Answer	Marks
8(a)	**1 mark** for each correct licence	**3**

Statement	Software Licence
A fee is charged for the software. The source code cannot be accessed. Users do not get a free trial.	**Commercial**

	Users can try the software before buying it. Users may not be able to access all features during the trial.	**Shareware**	
	Software is usually free of charge. Users can access the source code and alter the program to their needs.	**Open Source** // **Free software**	
8(b)	**1 mark** for each bullet point (**max 3**) e.g. • The professional ethical body has ethical guidelines to follow • … so clients/other staff know the standards being applied • … so the computer programmer does not have to decide what is ethical it's written down • Clients/staff know the computer programmer is reputable • … there may be a test/requirement for entry • … giving recognition of the computer programmer's skills/knowledge • The professional ethical body provides help and support • … for example, if the computer programmer needs legal advice • The professional ethical body runs training courses • … to keep the computer programmer's skills up to date		3

Question 9 · Oct/Nov 2025 Paper 12, original Q2 (a),(b) [6 marks]

Question	Answer	Marks
9(a)	**1 mark** for each correct row, **max 4** marks	**4**

Term	Description

	Copyright	Gives the holder the legal right of ownership of intellectual property
	Open Source (Initiative)	Software can be freely copied, distributed or adapted
	Shareware	Gives the user a trial period / limited functionality before a payment/sign up may be required
	Software Licence	A (legal) agreement that defines the rights and terms of usage for the software

9(b)	**1 mark** per bullet point, **max 2** marks **e.g.** • to create a safe, respectable and professional working environment for all employees • to ensure that employees understand the expectations • ... and the consequences of their actions • to provide a clear outline of what is considered good practice • so there are clear rules to follow in certain situations • to protect the reputation of the organisation // maintain public trust	**2**

Question 10 · Oct/Nov 2025 Paper 12, original Q10 (b),(c) [3 marks]

Question	Answer	Marks

Total: 59 marks

10(a)	**1 mark** per bullet point, **max 2** • The software is free of restrictions • The software is not necessarily free of charge • The source code comes with the program // A user can edit the source code to add functionality / suit their needs • Users can share the software with others	**2**
10(b)	**1 mark** for a correct answer • Digital Signature	**1**

Question 11 · Oct/Nov 2025 Paper 13, original Q7 (a) [2 marks]

Question	Answer	Marks

11(a)	**1 mark** per point, **max 2** marks e.g. • The learning experience could be improved • … by identifying students who are struggling • …. and employing more personalised learning • There could be early Intervention • … to determine which students require extra support • • Teachers and students could understand the challenges better • …and identify optimal times for learning • There could be privacy concerns • …. students / teachers could be uncomfortable with constant surveillance • ... students might not want their data / actions passed to third parties • There could be Mental Health concerns • …. due to constant monitoring and pressure • …. contributing to student stress and anxiety	**2**

Question 12 · Oct/Nov 2025 Paper 13, original Q8 (c) [2 marks]

Question	Answer	Marks
12(a)	**1 mark** per bullet point, **max 2** marks • Open Source (Initiative) • Free Software (Foundation)	**2**